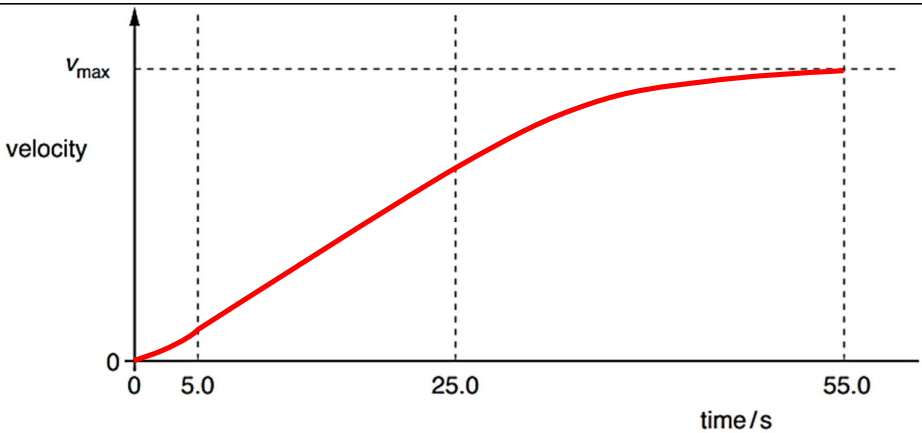


1(a)(i)	Product of mass and velocity.	[1]
(a)(ii)	Impulse = $\Delta(mv) = m \Delta v$ $= 1000 (0 - 80 \times 1000 / 3600)$ $= (-) 2 \times 10^4 \text{ kg m s}^{-1} \text{ or N s}$ (accept: 800-2000 kg for mass, 70-100 km/h for top speed) [1] impulse = change in momentum [1] correct estimates for mass & top speed (no need 1 sf) [1] correct unit & final value (show <u>conversion to m s⁻¹</u>)	[1] [2]
(b)(i)	Any two from the following: air resistance <u>varies</u> (with speed) friction (with ground / runway / tarmac) <u>varies</u> thrust / (resultant) force <u>varies</u> mass of fuel <u>decreases</u> / fuel used up	[2]
(b)(ii)	Area under graph = change in momentum $(\frac{1}{2} \times 5 \times 1.13 \times 10^5) + (20 \times 1.13 \times 10^5) + (30 \times 1.05 \times 10^5)$ $= 5.69 \times 10^6 \text{ (N s)}$ $v = p / m = 5.69 \times 10^6 / 7.31 \times 10^4 = 77.9 \text{ m s}^{-1}$ (allow ECF for wrong area)	[1] [1] [1]
(b)(iii)	 [1] curve of increasing gradient from origin [1] line of constant gradient [1] decreasing positive gradient after 25.0 s Deduct [1] if graph is not smooth or has kinks Note: The end point of graph should not be horizontal	[3]
(b)(iv)	Distance = area under graph $= \underline{2000 \text{ to } 3200 \text{ m}}$	[1] [1]